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School of Social Sciences
Indira Gandhi National Open University

MEC-203

QUANTITATIVE METHODS

Volume-2

MEC-203

QUANTITATIVE METHODS
(Vol. 2)

School of Social Sciences
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Maidan Garhi, New Delhi-110068

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PRINT PRODUCTION

Mr. Tilak Raj, Assistant Registrar, IGNOU, New Delhi

SECRETARIAL ASSISTANCE

Mr. Ankur Gupta, SOSS, IGNOU, New Delhi

Graphics/ Cover Design – Mr. Lalit Kumar

September, 2023

ISBN : 978-93-5568-992-4

Further information on the Indira Gandhi National Open University courses may be obtained from the University's office at Maidan Garhi, New Delhi -110068 or visit our website: http://www.ignou.ac.in

Printed and published on behalf of the Indira Gandhi National Open University, New Delhi, by Registrar, MPDD.

Printed at : Dee Kay Printers, 5/37 A, Kirti Nagar Indl. Area, New Delhi - 110 015

COURSE PREPARATION TEAM

Unit	Block/Unit Title	Unit Writer
BLOCK 1	**REVIEW OF BASIC MATHEMATICS**	
Unit 1	Fundamental Concepts of Mathematics	Dr. Meenakshi Sridhar, Associate Professor, Department of Computer Sciences, Rajdhani College, University of Delhi, Delhi
Unit 2	Overview of Basic Methods of Mathematics	
Unit 3	Relations and Functions	
Unit 4	Co-ordinate Geometry and Representation of Functions	Prof. Gopinath Pradhan
BLOCK 2	**LINEAR ALGEBRA**	
Unit 5	Matrix Algebra	Sunando Basu
Unit 6	Vector Analysis	
Unit 7	Vectors and Matrices	
Unit 8	Vector and Matrix Representations of Linear Equations	Dr. Meenakshi Sridhar, Associate Professor, Department of Computer Sciences, Rajdhani College, University of Delhi, Delhi
BLOCK 3	**CALCULUS**	
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Unit 11	Differential Calculus: Functions of Several Variables	
Unit 12	Integration: Introduction and Techniques	Prof. Gopinath Pradhan
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Unit 17	Unconstrained and Constrained Optimisation	Rittwik Chatterjee
Unit 18	Advanced Topics in Optimisation– I	Dr. Ram Prasad Yadav, Assistant Professor, Department of Mathematics, SRM University, Sonepat, Haryana
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Unit 30	Estimation	Biswadeep Basu (updation by Ms. Nivedita Mullick)
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COURSE CONTENTS
(Volume 2)

MATHEMATICAL SYMBOLS/ NOTATIONS USED IN THE COURSE

Symbol	**Symbol Name/Meaning Attached**
[a,b]	Closed interval
(a,b)	Open interval
$\in$	Element of
Δ	Delta/Change in
dx	Differential
$\lim\limits_{\Delta x \to 0} f(x)$	Limit
ln	Natural log
$\int_a^b f(x)\,dx$	Definite Integral
$\forall$	Universal quantifier: :for all"
$\exists$	Existential quantifier: "there exists"
$\exists_1$	Existential quantifier: "there exists exactly one"
$\exists_{>1}$	Existential quantifier: "there exists more than one"
$\neg$	Negation; $\neg$P "not P"
$\wedge$	Conjunction: "and"
$\vee$	Disjunction: "or"
$\Rightarrow$	Implication; $p \Rightarrow q$"if p then q"
$\Leftrightarrow$	Equivalence: "if and only if"
$a, x \in \mathsf{R}$	Numbers, scalars
$a = (a_1, a_2, ..., a_n) \in \mathsf{R}^n_+$	Vector of parameters
$x = (x_1, x_2, ..., x_n) \in \mathsf{R}^n_+$	Vector of variables
A, X	Sets
A, X	Matrices (with dimensions *m* by) with *m* rows and *n* columns
$x \sim y$	Vector **x** is indifferent with respect to vector **y**
$\succ$	Strong preference relation
$\mathsf{x} \succ \mathsf{y}$	Vector **x** is strongly preferred over vector **y**
$\succsim$	(weak) Preference relation
$\mathsf{x} \succsim \mathsf{y}$	Vector **x** is (weakly) preferred over vector **y**
R	Set of real numbers
R_+	Set of nonnegative real numbers
$\text{int}\,\mathsf{R}_+$	Set of positive real numbers
$\mathsf{R}^n = \mathsf{R} \times \mathsf{R} \times ... \times \mathsf{R}$	$\mathsf{n}-$Dimensional space of real numbers, Cartesian product of set R
$\mathsf{R}^n_+ = \{x \in \mathsf{R}^n \mid x \geq 0\} \subset \mathsf{R}^n$	Nonnegative orthant (subspace) of set R^n
$\text{int}\,\mathsf{R}^n_+ \subset \mathsf{R}^n_+$	Interior of set R^n
$\mathsf{f} : \mathsf{X} \to \mathsf{Y}$	Function, $\mathsf{X}-$Domain of a function (set of arguments of a function)

	Y – Codomain of a function (set of values of a function)
$y = f(x)$	Scalar function of one variable x – Independent variable (argument/input of a function) y – Independent variable (value of function)
$y = f(x_1, x_2)$	Scalar function of two variables x_1, x_2 – Independent variables (arguments/inputs of a function) y – Independent variable (value of function)
$\frac{dy}{dx}, y', \ f'(x)$	Derivative of first order of function y = f(x)
$\frac{d^2y}{dx^2}, y'', \ f''(x)$	Derivative of second order of function y = f(x)
$\frac{\partial y}{\partial x_i}, \frac{\partial f(x_1, x_2)}{\partial x_i}$	Partial derivatives of first order for function $y = f(x_1, x_2)$ with respect to variable x_i $i = 1,2$
$\frac{\partial^2 f(x_1, x_2)}{\partial x_i \partial x_j}, \frac{\partial^2 (x_1, x_2)}{\partial x_i^2}, \quad i = 1,2$	Partial derivatives of second order for function $y = f(x_1, x_2)$
$H(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f(x_1, x_2)}{\partial x_1^2} & \frac{\partial^2 f(x_1, x_2)}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f(x_1, x_2)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x_1, x_2)}{\partial x_2^2} \end{bmatrix}$	Hessian, symmetric matrix of partial derivatives of 2nd order for function $y = f(x_1, x_2)$
$\left. \frac{df(x)}{dx} \right\|_{x=\bar{x}}$	Value of a derivative of one variable function at point $\bar{x}$
$\left. \frac{\partial f(x_1, x_2)}{\partial x_i} \right\|_{\mathbf{x}=\bar{\mathbf{x}}}$	Value of a derivative of one variable function at point $\bar{\mathbf{x}} = (x_1, \bar{x}_2)$
$\langle \mathbf{p}, \mathbf{x} \rangle = \sum_{i=1}^{2} p_i x_i$	Scalar product of two vectors $\mathbf{p}, \mathbf{x} \in R_+^2$
$\det(\mathbf{A})$ or $\|\mathbf{A}\|$	Determinant of matrix **A**
$d_E(\mathbf{x}^1, \mathbf{x}^2) = \left(\sum_{i=1}^{2} \left(x_i^1 - x_i^2 \right)^2 \right)^{\frac{1}{2}}$	Euclidean metric (distance)
$d(x^1, x^2) = \max_{i=1,2} \left\{ \left\| x_i^1 - x_i^2 \right\| \right\}$	Non-Euclidean metric (distance)
$\left\\| \mathbf{x}_E \right\\| = \left(\sum_{i=1}^{2} (x_i)^2 \right)^{\frac{1}{2}}$	Euclidean norm
$\left\\| \mathbf{x} \right\\| = \max_{i=1,2} \left\{ \left\| x_i \right\| \right\}$	Non-Euclidean norm
$t = 0, 1, 2, \ldots$	Time as discrete variable
$t \in [0; +\infty)$	Time as continuous variable
∇f, gradf, Gradient of function f, $\nabla f = (\partial f \partial x1, \ldots, \partial f \partial xn)$	

SOME LETTERS OF THE GREEK ALPHABET (AND CORRESPONDING SYMBOLS) YOU MAY ENCOUNTER

Symbol	Greek Alphabet	Symbol	Greek Alphabet
α	Alpha	θ	Theta
β	Beta	λ	Lamba
γ	Gamma	π	Pi
δ	Delta	σ	Sigma
ε	Epsilon	χ	Chi
ψ	Psi	μ	Mu
ρ	Rho	ω	Omega

BLOCK 4

REAL ANALYSIS